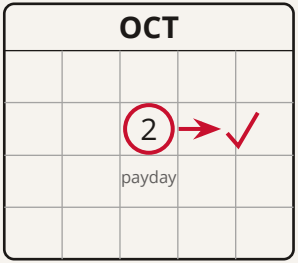
# Kept

Promises customers can keep: one guarded action, due on payday, released by a person.




Team Sharpe Shooters
CIBC Collections Hackathon · 16 October 2026

THE PROBLEM

# Collections today asks for money on the wrong day

- **Five systems, five keys per person.** Each files the same customer differently, so nobody sees the whole person.
- **Promises are dated by the calendar, not by pay.** A promise due before payday breaks; the bank's own data shows promised money paid is **+6.1 pts** higher when the due date falls within two days of payday.
- **More contact doesn't help.** The bank's randomised champion, two challengers and a hold-out keep promises at the same rate.
- **Hardship gets missed** in notes and calls, so vulnerable customers are pushed like anyone.

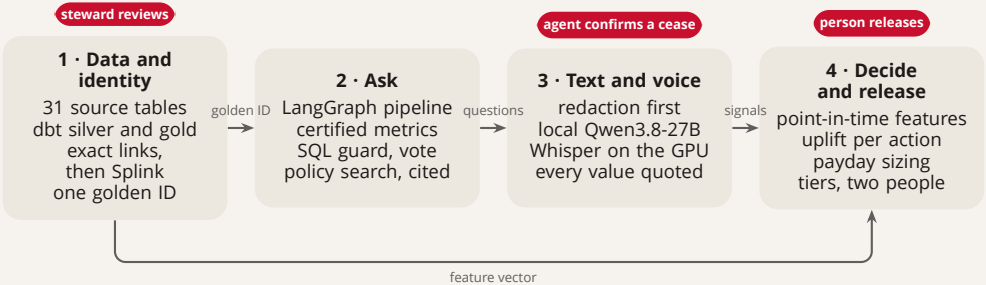
96,000
open cases tonight

$68.7M
balance at risk

38,115
need a hardship specialist

SYSTEM DESIGN

# Four layers, one guarded decision




Nightly: dbt build → features → score 96,000 cards (~85 s). Live: the API rechecks every release on fresh data.



Source: src/kept/cli.py (nightly), docs/design/SharpeShooters_SystemDesign.pdf

ARCHITECTURE

# A small stack that maps onto a bank's

| Piece | Here | At the bank |
|---|---|---|
| Warehouse | DuckDB file, dbt silver/gold, Feast online store; 20x–40x today's data on 4 GB, never failed | the bank's lakehouse; dbt models carry over |
| Audit | DuckDB, append-only, SHA-256 chain, `kept audit-verify`; audit page 7.9 s → 0.04 s | WORM storage, same chain |
| Models | LightGBM uplift + promise, MLflow, gate; uplift Qini 7.9 vs 3.2 for a response model; promise AUC 0.697 vs 0.625 | model risk management sign-off |
| Local language models | Qwen3.8-27B on llama.cpp, Whisper, bge-m3; SetFit reads 760,594 notes in 63 s, the LLM would take 29 days | an on-prem inference cluster |
| The Ask's model | the same local Qwen3.8-27B, no hosted API or keys; 86 of 93 right in the pipeline, 36 with one prompt | the bank's approved model |
| API and app | FastAPI; Next.js workspace; personas; portfolio overview ~40 ms | SSO, the bank's design system |

Speed: 96,000 cards in ~85 s; portfolio overview ~40 ms; audit page 7.9 s → 0.04 s after moving filters into SQL. 1,322 tests; every change through CI.



Source: README.md, pyproject.toml, docs/reports/decisions.md

LAYER 1 · DATA AND IDENTITY

# Five keys become one person, and a steward checks the doubtful ones

- **Exact links first** (shared account and customer numbers), **then probabilistic matching** (Splink, names, dates of birth, addresses) for the rest.
- **Uncertain pairs go to a person**: 36,279 wait in the steward queue, oldest first, before two records ever merge.
- **Data quality is a contract**: 31 tables checked on load (types, ranges, duplicates), issues reported, not hidden.

99%
of 200 injected duplicates found

0
wrong merges

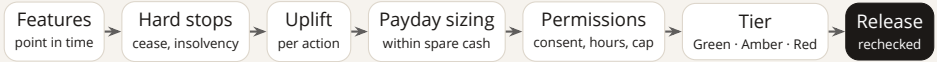
36,279
pairs for a steward, paged



Source: docs/reports/identity_recall.md, QUALITY_REPORT.pdf

LAYER 4 · DECIDE AND RELEASE

# From a feature vector to a released action, in seven checked steps




**Uplift, not propensity**

LightGBM T-learner: one model per action against waiting. Contact whoever a contact changes, not whoever pays anyway.

**Due on payday**

Next payday rolled forward from deposits or asked on the pay page; the promise falls due that day, the measured peak, inside spare cash.

**People release**

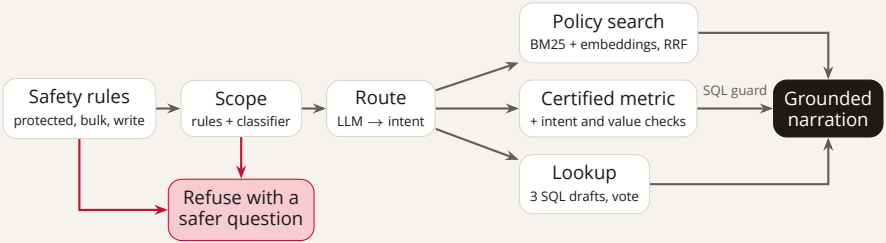
Green in a sampled team-lead batch; Amber by the agent; Red by a hardship specialist and a second person. Every release rechecks on fresh data.



Source: src/kept/nba/engine.py, src/kept/nba/release.py

LAYER 2 · THE ASK, AS A LANGGRAPH

# Every answer walks a graph, and shows the path it took




Every number in an answer must appear in the rows it came from; the SQL guard allows read-only queries on approved tables and columns only; the trace (*Route: metric · SQL guard: allowed · 1 row*) shows under each answer.

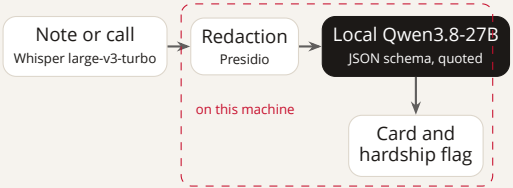
**On the local model: released 21/21** · stress suite 38/42 · unseen drill 27/30. Two deterministic checks fix what the router can slip on: a channel the question names, a value the warehouse stores differently.



Source: src/kept/ask/graph.py, docs/reports/ask_accuracy.md

LAYER 3 · TEXT AND VOICE

# Hardship heard the first time, without the words leaving the machine




- Every extracted value carries the exact quote it rests on; agents confirm a cease before it stops contact.
- Replies are decoded to the JSON schema: malformed replies went from 152 of 1,172 calls to 0.

**1.00**
hardship recall, notes and calls

**9.4%**
word errors (bank's speech-to-text: 12.4%)

**≥0.80**
F1 on every extracted field

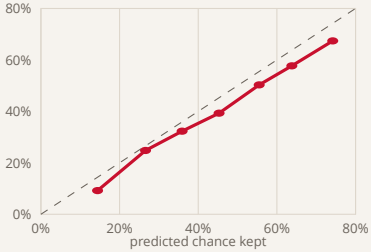


Source: docs/reports/extraction_gate.md, voice_wer.md

EXPLAINABILITY

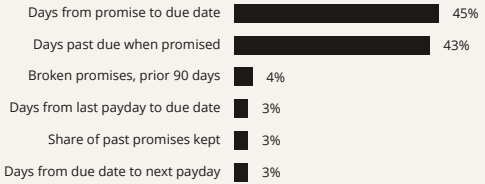
# Every card says why, and the models say how far to trust them

## Promise model: is a 60% promise kept 60% of the time?




AUC 0.610, Brier 0.221 on 50,653 promises it never saw (isotonic calibration). Dashed: perfect.

## What drives it (share of |TreeSHAP|)




Per card: the 12 largest TreeSHAP contributions, both directions, on the case page. Uplift model: Qini 0.0008 ± 0.0004 on 141,350 held-out decisions.



Source: data/exports/reports/model_card.json (scripts/model_card.py)

DECISIONS, AND THE ALTERNATIVE WE REJECTED

# Why it's built this way

| We chose | Instead of | Because |
|---|---|---|
| Uplift per action | a propensity-to-pay score | top 30%: 122 extra payers per 1,000 against 81 (difference 26 to 56); the score picks people who pay anyway (56% against 26.2%) |
| Timing (payday) | more channels or intensity | the bank's A/B: strategies keep promises at the same rate; payday timing adds +6.1 pts |
| Local models for everything | a hosted model | not cost (Haiku 5.5: USD 1,386 a year, ours 2,200); customer data stays inside (OSFI B-10, E-23, PIPEDA, Law 25) |
| Certified metrics first | free-form SQL from a model | router and certified metrics: 36 to 81 right of 93 (38.7% to 87.1%) |
| A graph with checks (LangGraph) | one prompt | 86 of 93 right against 36; wrong but confident 33 → 7 |
| Hardship by a rule; two people release Red | a model on the same cash-flow inputs | rule recall 0.652 vs model 0.672 at the same flag rate, no precision gain, and the model adds a gender flag |
| SetFit for note history | the LLM on every note | hardship F1 0.87 vs 0.88; 63 s for all notes vs about 29 days |
| Hybrid policy search (open) | dense only | dense MRR 0.747 vs 0.640 for hybrid (French 0.821 vs 0.564); hybrid still live |
| Schema-constrained decoding | parse-and-retry | malformed replies 152 → 0 of 1,172 calls |
| Qwen3.8-27B on llama.cpp | Gemma 3 12B on Ollama | Ollama ran a 4k window under a 9k prompt; released 18 → 21/21 |

RIGOUR

# What we tried, measured, and didn't ship

### A bigger local model

Gemma 4 26B on a 16 GB GPU: 24 s a note against 3 s, runaway replies, no better on held-out notes. Rejected; the dense Qwen3.8-27B fits and is faster.

### A tuned uplift model

40 candidates tuned on a validation window; the best lost to the champion on held-out decisions (Qini 0.0029 vs 0.0073). X- and DR-learners won some arms and lost others. The gate kept the champion.

### A leaky feature

amount_ratio made AUC 0.92: it encoded the outcome. Dropped; honest AUC 0.63. A wider model then matched the bank's (0.697 vs 0.702); averaged with it, 0.763.

The rule: a change ships only if it beats what's live by more than the noise, on data it never saw.

SAFE BY CONSTRUCTION

# Checked every night, not promised once

**0 / 96,000**
cards change when every protected attribute is scrambled and the book rescored

**0.87–1.00**
contact ratio across 11 protected attributes; a minimum-events rule

**1 chain**
every recommendation, release and override hash-chained; one edit breaks it

**1 switch**
reverts everyone to the bank's strategy; a worse model can't go live

- **Roles:** agents see their own cases; analysts see aggregates only (groups under 10 suppressed, no single case); names masked for every role but compliance.
- **Drift:** 8 features drift, 5 to watch, 13 stable (PSI vs training); 1 payroll features never appear in training rows. Drift on a model input pauses promotion until a retrain is checked.
- **Voice biometrics, accent and pitch are never inputs**; accent is used only to check Whisper works equally well (9.1–10.1% across regions).



Source: docs/reports/fairness_report.pdf, separated_review.md, drift_report.pdf; kept audit-verify

ONE CUSTOMER

# Meet Jane: her last promise fell due before payday, and broke

- Pay lands in 3 days and is down 27% on her average.
- She answers calls after 18:00, not in the morning slot the strategy uses.
- Her notes say her shifts were cut: a hardship signal nobody acted on.

**Kept's card for Jane**

Call, Tue 18:00–20:00
$140 a month × 3, due on payday, inside ~$160 spare cash

Why: pay lands in 3 days (+9 pts) · pay down 27% (+6) · answers after 18:00 (+4)

**Red** hardship: a specialist and a second person release it



Source: the guided demo (case CS-2026-118204) 

THE PRODUCT

# One workspace for every role

**Queue and case**
ranked by expected recovery; a cited case brief from the local model; the full "why"; contacts, promises and month-ends from the warehouse

**Portfolio**
tonight's book, the A/B as a forest plot, cohorts, insights with their SQL, models, fairness and drift

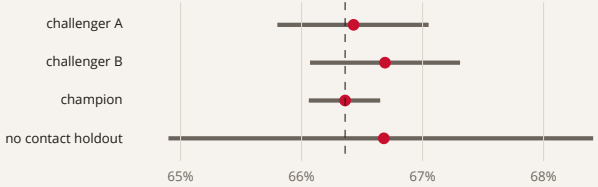
**Ask, approvals, audit**
plain-English questions with traced paths and sources; stage-by-stage release; the hash-chained trail

Notifications come from the audit log (nothing stored twice); the customer's pay page asks for payday when the bank doesn't know it (about 3 customers in 4).

MEASURED IMPACT

# The bank's own randomised test: strategy doesn't move promises, timing does

**Promise-kept rate by strategy (95% Wilson interval)**




Every interval overlaps the champion's (dashed).

## +6.1 pts

promised money paid when due 0–2 days after payday (95% CI 2.8–10.6), 12,808 promises in randomised cases

## +61 per 1,000

more customers paying within 14 days when Kept chooses the action, paired on the same decisions (95% CI 55–67; off-policy, doubly robust)



Source: silver.collections_cases (closed, randomised); evidence.md §1–2, §4

IS IT REAL?

# Every claim, its test, its 95% interval

| Claim | Test | Result [95% CI] | |
|---|---|---|---|
| Payday dating raises money paid | randomised cases | +6.1 pts [2.8–10.6] | holds |
| It is the payday, not the customer | vs 1–3 days either side | +7.7 [5.0–9.8] · +8.3 [6.1–10.5] | holds |
| Same customer, on vs off payday | fixed effects | +4.0 pts [1.1–7.4] | holds |
| Fake paydays show nothing | paydays moved 3–10 days | 6 of 6 intervals include 0 | holds |
| Promise model vs the bank's | paired bootstrap | 0.697 vs 0.702 [-1.9 to +1.0] | tie |
| The two models together | rank average, nothing fitted | 0.763, +6.1 [5.1–6.9] | holds |
| Kept's action vs the bank's | doubly robust, paired | +61 payers / 1,000 [55–67] | holds |
| Targeting by uplift | randomised cells only | $+18K to $+42K / 1,000, CI spans 0 | unproven |
| Pays back in year one | 20,000 simulated draws | 99.97% (98.5% if correlated) | holds |

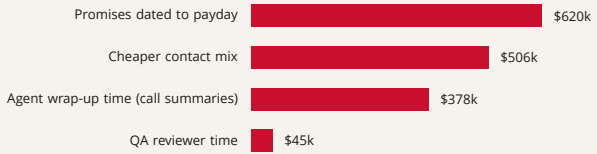


Source: evidence.md §2, §2a, §4, §8, §9; business_case.pdf

BUSINESS VALUE

# $1.55M a year in the base case, and where it comes from

## Base case by driver ($k a year)




Payday dating: more promised money paid on the promises made each year. Contact mix: fewer calls, more payment links. Wrap-up: call summaries agents don't type. QA: reviewers check flagged calls only.

**$1.55M**
a year, base case (low $0.44M, high $4.27M)

**2.4×**
year-one ROI, base case

**2.6 months**
payback, base case

99.97% of 20,000 draws over every assumption pay back in year one (98.5% if they all move together). With no payday effect it still pays back; with everything else at its low end it needs 0.2 pts.


Source: docs/reports/business_case.pdf, data/exports/reports/business_data.json

VALUE BEYOND DOLLARS

# What the customer and the regulator get

| Guard (must not get worse) | Bank today | Kept |
|---|---|---|
| Contact-cap breaches | 0.03% of cases | 0 (checked at release) |
| Contact outside calling hours | 1.4% of cases | 0 |
| Hardship notes missed | not measured | recall 1.00 |
| Any-contact fairness ratio | not measured | 0.87 or above |
| Complaints | 3.1% of cases | needs a live pilot cell |

Every action is explained (fixed sentences, never model text), on the record, and released by a named person.



Source: evidence.md §5 (guard metrics)

NEXT

# A pilot that proves it, with a rule that ends it either way

- **20% hold-out first**: the payday effect shows within 2 weeks (80% power); then a permanent 5% on the bank's strategy (7 weeks to the same answer).
- **A randomised due-date cell**, so the payday effect becomes a trial, not an estimate.
- **Primary metric:** promised money paid. **Guards:** complaints, caps, hours, fairness ratio.

**Decision rule**

Stop early for harm if any guard worsens beyond its 95% interval; scale up if promised money paid beats the hold-out's interval for two consecutive months; otherwise run to the planned size.

**Day one**

kill switch, nightly drift and fairness checks, the audit chain, and payday asked where the bank doesn't know it.

HOW WE BUILT IT

# Four tracks, gated, every number reproducible

- Data · Ask · Text and GPU · Decisions, integrated at gates G0–G4.
- 170+ pull requests, each through CI; 1,322 tests.
- Local models only: no hosted API and no keys; nothing leaves the machine.
- `JUDGES.md`: every claim, its report, and the command that reproduces it.

**Reproduce any number**

```
uv run kept nightly
uv run python scripts/evidence_report.py
uv run python scripts/model_card.py
uv run python scripts/bench_run.py ...

bash scripts/secret_scan.sh
```

Jane’s promise falls due on her payday, inside what she can spare, and her hardship was heard the first time she said it.

APPENDIX

# Who can see and do what

| Role | Sees | Names | Releases | Ask |
|---|---|---|---|---|
| Collections agent | own cases | masked | Amber on own cases | yes |
| Hardship specialist | every case | masked | Red, first approval | yes |
| Team lead | every case | masked | Red second; Green batch | yes |
| Compliance | every case and the full trail | unmasked | Red second | yes |
| Analyst | aggregates only (groups ≥ 10) | none | none | yes |
| Data steward | match pairs | masked | merge or split | yes |



Source: src/kept/governance/access.py, src/kept/web/src/lib/session.ts

APPENDIX

# The Ask's benchmarks, on the local model

**21/21**
the organisers' released questions with dev gold

**38/42**
our stress suite, aimed at the release's traps

**27/30**
an unseen drill we never tuned on (one question in French)

- Every model call is local (Qwen3.8-27B on llama.cpp): no hosted API, no keys, no quota. The graph's deterministic checks (channel, stored values, column repair, dates, cohorts, relevance, "can we" means policy) and a convention linter keep the model on the rails; drafts that disagree escalate to five, then the model judges the results.
- A refusal must refuse; a number must match gold within the stated tolerance; a policy answer must cite the section.



Source: data/exports/reports/{released,stress,drill}_answers.json; docs/reports/ask_accuracy.md

APPENDIX

# The uplift search we ran, and why the champion stayed

| Model, on held-out decisions | Qini, calls | noise | Qini, any contact |
|---|---|---|---|
| Champion (live) | 0.0073 | 0.0003 | 0.0008 |
| Tuned challenger | 0.0029 | 0.0003 | -0.0006 |

40 candidates were compared on a validation window inside the training period (15 Jul to 15 Aug); only the best was judged once on the test decisions, by the nightly's own gate. It lost beyond noise, so the champion stays.

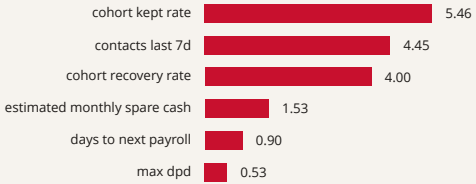


Source: docs/reports/tuning.md (scripts/tune_uplift.py)

APPENDIX

# Drift and fairness, in detail

## Largest population shifts (PSI, live vs training)




Above 0.25 is drift. The promise-kept rate fell from about 63% to 35% in 2026, so cohort rates moved most; drift on a model input pauses promotion.

## Fairness flags (four-fifths rule, ratio under 0.80)

gender code, call: 0.72; marital status, call: 0.73; marital status, hardship review: 0.76.

Each is reviewed: small groups, or explained by recorded hardship. Protected attributes are never model inputs, and scrambling them changes 0 of 96,000 cards.



Source: data/exports/reports/drift_data.json, fairness_data.json

APPENDIX

# Backend: one machine to 40x

| Median, same machine | DuckDB | Polars | PostgreSQL | PySpark |
|---|---|---|---|---|
| roll_rate 1x | 0.9 s | 0.5 s | 10.4 s | 5.3 s |
| roll_rate 10x | 9.8 s | out of memory | skipped (disk) | 38.9 s |
| silver build 1x | 11.4 s | 2.4 s | 9.5 s | 10.7 s |
| silver build 10x | 94.3 s | out of memory | skipped (disk) | 96.2 s |

8 GB memory cap; DuckDB limited to 4 GB and spilling to disk. DuckDB reached 40x on `roll_rate` (61.7 s) and extrapolates to about 42x before the 24 GB limit.

### Nightly batch, not hourly

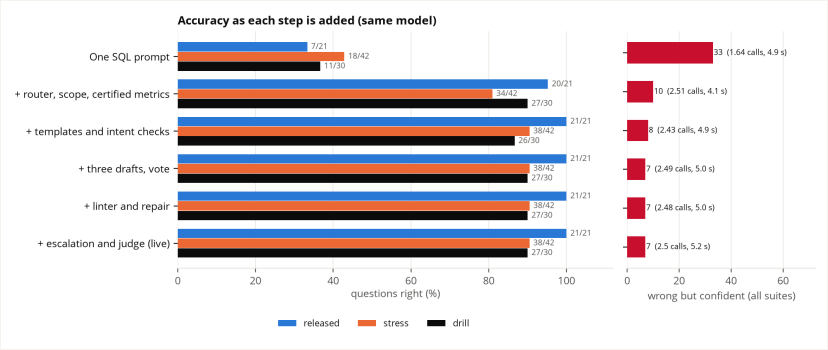
Re-scoring every hour changes 3.1% of cards by the end of a day (2,839 of 90,998); 2.7% of cards change with no new note, which the live-note path does not cover.

### One feature definition

A rewrite with the wrong clock differs on 1,325 of 56,000 cells, a wrong time zone on 808; the control on 0. `kept parity` stops both; the clock bug would change 5,503 cards.

APPENDIX

# The Ask: the router and certified metrics carry the gain




Same model, 93 graded questions. Router and certified metrics: 38.7% to 87.1%, wrong-but-confident answers 33 to 10.

Templates add 4; drafts, vote, linter and judge add 1.



Source: docs/reports/decisions.md B5; data/exports/reports/decision_B5.json

APPENDIX

# Three model choices, measured

| Choice | Option | Score | Also |
|---|---|---|---|
| Reading hardship in notes (B7): F1; hours for 760,594 notes | Keyword rules | 0.63 | recall 0.75; 0.003 h |
| | SetFit (GPU) | 0.87 | recall 0.98; 0.018 h |
| | LLM | 0.88 | recall 1.00; 701 h (extrapolated) |
| Ranking contacts (B10): Qini in noise units; extra payers per 1,000 in the top 30% | Response model | 3.2 | 81 per 1,000 |
| | T-learner (live) | 7.9 | 122 per 1,000 |
| | S-learner | 14.8 | 146 per 1,000 (open: pilot decides) |
| | Causal forest | 10.7 | 134 per 1,000 |
| Promise model (B9): AUC | Six features | 0.625 | easy to explain |
| | Wide model | 0.697 | the bank's score: 0.702 |



Source: docs/reports/decisions.md B7, B9, B10

APPENDIX

# Which local model

| Model | GPU MiB | Ask right, of 93 | Confident wrong | Notes F1 | Brief, s |
|---|---|---|---|---|---|
| Qwen3.8-27B (live) | 14,733 | 86 | 7 | 0.88 | 4.79 |
| Qwen3-14B | 12,802 | 74 | 13 | 0.905 | 2.92 |
| Qwen3-8B | 8,546 | 75 | 17 | 0.87 | 1.54 |
| Gemma 3 12B | 10,520 | 71 | 16 | 0.739 | not run |

Only one model fits the 16,303 MiB card, so it serves all three jobs. Gemma 3 12B was stopped after it failed the notes hardship gate. Ollama (same Qwen3-14B file) cut a 9,743-token prompt to 4,096 by default; with the context matched it runs at 78.3 tokens a second against 79.4 on llama.cpp. Live red team: 41 of 43 attacks refused; the other 2 answered only the legitimate part.



Source: docs/reports/decisions.md B1, B3; data/exports/reports/decision_B1.json, decision_B3.json